

VLM-Guided Hierarchy-Aware Prompt Learning for Remote Sensing Object Detection

Research Group
Department / Laboratory Placeholder
contact@example.com

Abstract

Remote-sensing object detection requires recognizing fine-grained categories in large, cluttered, and strongly oriented aerial scenes. Recent open-prompt detection work has shown that language supervision can help detectors generalize beyond closed label sets, while 2026 vision-language remote-sensing studies point to a newer direction: multimodal prompting and VLM-assisted label purification can reduce the semantic noise that appears when class names, scene context, and weak labels disagree. This short method paper reframes our GeoNexus-RSD/OpenPrompt scaffold as a VLM-guided, hierarchy-aware prompt learning framework for DOTA-style rotated object detection. The proposed system combines a hierarchy-aware prompt bank, a scene-context prompt adapter, and VLM-assisted pseudo-label purification. OpenPrompt is used as the local implementation scaffold rather than the main novelty. Optional Gumbel routing is retained only as a lightweight extension for branch selection. Because final experiments are not yet complete, we present the motivation, method, planned evaluation protocol, and expected contributions without claiming measured state-of-the-art performance.

1 Introduction

Remote-sensing imagery differs from everyday natural-image detection in ways that make prompt learning both attractive and difficult. Objects are often small relative to the image, densely packed, partially occluded, and captured from near-nadir viewpoints. Rotation is not an exception but a central property: airplanes, ships, bridges, vehicles, storage tanks, and sports fields can appear at arbitrary angles. At the same time, many operational class vocabularies are fine-grained and hierarchical. A detector may need to distinguish small vehicle from large vehicle, ship from harbor equipment, and baseball diamond from ground track field while also remaining robust to broader parent concepts such as vehicle, vessel, and sports facility.

DOTA-style rotated object detection is therefore a natural target for a prompt-based remote-sensing method. The task stresses geometric localization, class disambiguation, and dense-scene reasoning. However, a purely text-prompted detector can be brittle when class prompts are shallow, when visual context conflicts with the object crop, or when weak labels contain systematic noise. A class name such as “small vehicle” is not enough to describe the spatial, contextual, and scale cues that separate it from nearby categories. Likewise, a generated pseudo label can look plausible at the category level while still violating scene evidence or taxonomy constraints.

Recent work suggests a more robust path. OpenRSD introduced the goal of open prompts for object detection in remote-sensing images and provides an important prior for prompt-conditioned rotated detection [1]. RS-MPOD studies multimodal prompting for remote-sensing object detection, indicating that prompt design should incorporate more than a flat text label [2]. SemiEarth demonstrates that vision-language models can help purify pseudo labels in semi-supervised remote-sensing segmentation, offering a practical mechanism for filtering weak supervision before it harms training [3]. These directions motivate a 2026-facing GeoNexus-RSD story: VLM-guided prompt learning should be hierarchy-aware, context-aware, and conservative about noisy pseudo supervision.

This paper describes a proposed method rather than a completed benchmark report. The local OpenPrompt repository supplies reusable parts: a DOTA-style data path, a prompt-bank scaffold, an OpenRSD-like detector interface, and GeoNexus-RSD modules for hierarchy and context. The intended contribution is not to claim that this scaffold alone solves open-vocabulary detection. Instead, we propose a defensible experimental program centered on three components: a hierarchy-aware prompt bank for fine-grained remote-sensing classes, a scene-context prompt adapter for small or confusing objects, and VLM-assisted pseudo-label purification for semi-supervised or weakly supervised expansion.

2 Related Work

Open-vocabulary vision and prompt learning. CLIP established that large-scale image-text contrastive pretraining can transfer visual recognition to novel textual categories [4]. In detection and segmentation, this idea is usually adapted through prompt embeddings, class-name templates, region-text alignment, or lightweight tuning. For remote sensing, direct transfer is challenging because aerial objects are observed at unusual scales and viewpoints. A prompt such as “a photo of a ship” may encode useful semantic priors but may not encode the difference between ship, harbor, bridge, and background clutter in a dense satellite tile. Prompt learning for this domain should therefore encode class hierarchy, aliases, and contextual contrasts instead of relying only on generic templates.

Remote-sensing open-prompt detection. OpenRSD is the most direct prior for this project. It formulates open prompts for remote-sensing object detection and gives the local project a useful baseline direction: prompt-conditioned detection over aerial scenes with rotated boxes [1]. Our framing differs in emphasis. OpenRSD and the local OpenPrompt scaffold are treated as starting infrastructure. The main method story is the combination of hierarchy-aware prompt construction, scene-conditioned adaptation, and VLM-guided purification. This avoids making the implementation scaffold itself the novelty.

Multimodal prompting and VLM purification. RS-MPOD, a 2026 arXiv work on multimodal prompting for remote-sensing object detection, supports the view that remote-sensing detectors benefit from prompts that combine semantic and visual information rather than isolated class strings [2]. SemiEarth, another 2026 arXiv work, uses VLM-purified pseudo labels for semi-supervised remote-sensing semantic segmentation and reports released code [3]. Although segmentation and detection have different output structures, the purification principle is transferable: before weak labels are used for training, a VLM can provide semantic consistency checks, scene-level plausibility checks, and class-confusion filtering. In rotated detection, this becomes especially useful for small objects and ambiguous categories where noisy boxes are common.

Parameter-efficient and discrete adaptation. LoRA provides a practical way to adapt large pretrained components through low-rank updates rather than full fine-tuning [5]. This is relevant when adapting text encoders, cross-modal projection layers, or prompt adapters under limited compute. Gumbel-Softmax provides differentiable sampling for categorical routing decisions [6, 7]. In this paper, however, routing is deliberately secondary. A Gumbel router can be tested as a lightweight extension for choosing among prompt branches or fusion heads, but the main claim should remain the VLM-guided hierarchy-aware prompt pipeline.

3 Method

3.1 Problem Setup

The target task is rotated object detection on large aerial images, with DOTA-style annotations represented as oriented bounding boxes and class labels. Let an input tile be I , a detector produce rotated boxes $b_i = (x_i, y_i, w_i, h_i, \theta_i)$, and a class vocabulary be organized as a taxonomy $\mathcal{T}$. Each leaf class c has parent categories, aliases, and confusion sets. The detector should produce class scores and rotated boxes while remaining robust to class names not seen during the initial supervised training split or to prompt variants that use parent-level or synonym-level descriptions.

The proposed training regime has three supervision sources. First, labeled DOTA-style boxes provide the supervised base. Second, prompt text provides semantic conditioning through a prompt bank. Third, unlabeled or weakly labeled tiles can provide candidate pseudo boxes, but only after VLM-assisted purification. This staged design reduces the risk of turning a noisy prompt expansion process into noisy detector supervision.

3.2 Hierarchy-Aware Prompt Bank

For each class c , we construct prompts at several levels of specificity. A leaf prompt names the target class directly, such as “small vehicle in an aerial image”. A parent prompt connects the class to a broader group, such as “road vehicle”. A contrast prompt states a visual distinction, such as “small vehicle, not large truck or building rooftop”. A context prompt adds likely scene evidence, such as “small vehicle on a road, parking lot, or airport apron”. These prompts form a set $\mathcal{P}_c = \{p_{c,1}, \dots, p_{c,K}\}$.

A text encoder maps each prompt into an embedding $e_{c,k}$. The class embedding is an aggregation of prompt embeddings,

$$\bar{e}_c = \sum_{k=1}^K \alpha_{c,k} e_{c,k}, \quad (1)$$

where $\alpha_{c,k}$ can be fixed, learned, or scene-conditioned. Hierarchical consistency is encouraged by smoothing related classes in the taxonomy. For example, child-class scores should not strongly contradict parent-class evidence unless the detector has high confidence in the local object cue. This is not intended to replace box-level supervision; it is a semantic regularizer that keeps prompt predictions aligned with the taxonomy.

The prompt bank also stores aliases and negative contrasts. Remote-sensing labels often have dataset-specific naming conventions, and a model that only sees one label string can overfit to phrasing. Alias prompts improve robustness to vocabulary variation, while contrast prompts focus the model on known failure modes such as tennis court versus ground track field, bridge versus harbor structure, and small vehicle versus storage tank.

3.3 Scene-Context Prompt Adapter

Small and confusing objects are rarely recognized from object appearance alone. A small bright rectangle may be a vehicle, a container, or a rooftop element depending on its neighborhood. The scene-context prompt adapter therefore modulates prompt embeddings using tile-level and region-level visual context. Given a visual feature map $F(I)$ and a pooled scene descriptor g , the adapter predicts a prompt offset or prompt weight:

$$\tilde{e}_c = \bar{e}_c + A(g, s_c), \quad (2)$$

where A is a lightweight adapter and s_c contains optional class statistics such as expected scale range, aspect-ratio range, and parent category. In a parameter-efficient version, A can use LoRA-style low-rank updates applied to projection layers, limiting the number of trainable parameters while allowing the prompt representation to adjust to aerial context.

The adapter is especially useful for rotated detection because geometric cues are part of the class evidence. Long thin objects on water are more likely to be ships or bridges depending on surrounding texture. Compact objects near runways may be airplanes or vehicles depending on size and arrangement. The adapter does not hard-code these rules; it gives the model a path to condition prompt embeddings on scene evidence before matching them with region features.

3.4 VLM-Assisted Pseudo-Label Purification

Pseudo labels can expand training data, but remote-sensing pseudo labels are often noisy. Dense scenes create duplicate boxes, small objects are missed, and visually similar classes are confused. Inspired by SemiEarth’s VLM-purified semi-supervised learning direction [3], we introduce a purification stage before pseudo boxes are accepted for detector training.

The purification stage has three checks. First, a semantic agreement check compares the candidate class with VLM responses to object crops and local context windows. The goal is not to trust the VLM blindly, but to reject candidates where text, crop, and context are inconsistent. Second, a hierarchy check maps the candidate label to its parent categories and rejects predictions that conflict with high-confidence parent-level evidence. Third, a geometry-context check filters boxes whose scale, aspect ratio, or orientation is implausible for the proposed class under the current scene context.

Accepted pseudo labels are assigned soft weights rather than treated as ground truth. If q_i denotes the purification confidence for pseudo box i , the pseudo-label loss is weighted as

$$\mathcal{L}_{pseudo} = \sum_i q_i (\mathcal{L}_{cls}(i) + \lambda_{box} \mathcal{L}_{box}(i)). \quad (3)$$

This conservative weighting is important because VLMs may not be reliable for every aerial category. The purification module should improve the supervision pool by removing obvious conflicts, not by manufacturing unsupported labels.

3.5 Detector Integration

The detector follows an OpenRSD-like prompt-conditioned rotated detection interface. Visual features are extracted from tiled aerial images. Region or query features are aligned with the adapted class prompt embeddings, and the detection head predicts class scores and oriented boxes. The implementation can be built inside the local OpenPrompt scaffold, which already contains DOTA-style loading utilities, prompt-bank construction, and configurable GeoNexus-RSD modules.

Training uses a supervised detection loss on labeled data and a weighted pseudo-label loss on purified unlabeled data:

$$\mathcal{L} = \mathcal{L}_{det} + \lambda_h \mathcal{L}_{hier} + \lambda_p \mathcal{L}_{pseudo}. \quad (4)$$

Here $\mathcal{L}_{hier}$ penalizes hierarchy-inconsistent predictions, while $\mathcal{L}_{pseudo}$ is enabled only for accepted pseudo labels. The model should first be validated in a fully supervised setting before semi-supervised purification is introduced.

Table 1: Planned ablation matrix. Cells are intentionally left as protocol placeholders because final experiments have not yet been completed.

Setting	Hierarchy prompts	Context adapter	VLM purification	Optional routing
Baseline detector	–	–	–	–
Prompt bank only	yes	–	–	–
Context-aware prompts	yes	yes	–	–
Purified semi-supervision	yes	yes	yes	–
Routing extension	yes	yes	yes	soft / Gumbel

3.6 Optional Gumbel Routing

The local codebase contains routing hooks, including Gumbel-Softmax. In this paper, routing is not presented as the central novelty. It is an optional extension that can choose among prompt branches, adapter strengths, or fusion heads. For example, a route variable z may select a fine-grained prompt branch for dense vehicle scenes and a context-heavy branch for harbor scenes. A Gumbel-Softmax relaxation allows this choice to be trained differentiably. The planned ablation should compare no routing, soft routing, Gumbel routing, and random routing to determine whether discrete branch selection adds value beyond the hierarchy and VLM purification components.

4 Experimental Protocol

4.1 Datasets and Tasks

The primary evaluation will use DOTA-style rotated object detection. Images are split into manageable tiles, and annotations are converted to oriented boxes. The method should be evaluated under standard supervised training and under a semi-supervised extension where additional unlabeled or weakly labeled tiles are purified before use. If cross-dataset testing is available, the same prompt bank can be evaluated on held-out vocabularies or prompt variants to measure vocabulary robustness.

The main metrics are rotated mean average precision, class-wise AP, and confusion statistics for fine-grained categories. Because prompt methods are sensitive to wording, we also plan a prompt-robustness evaluation that compares exact class labels, alias prompts, parent-level prompts, and mixed prompt sets. Runtime and memory should be reported because prompt banks and VLM purification can introduce nontrivial overhead.

4.2 Baselines

The first baseline is the local OpenPrompt/OpenRSD-style detector without the proposed hierarchy adapter or VLM purification. The second baseline adds only the hierarchy-aware prompt bank. The third adds scene-context prompt adaptation. The fourth adds VLM-assisted pseudo-label purification. Optional routing is evaluated after these components are stable. This order is important: routing should not be used to hide instability in the core prompt and purification pipeline.

4.3 Ablations and Analysis

The planned ablations isolate the following questions. First, does hierarchy-aware prompting improve fine-grained class discrimination compared with flat class-name prompts? Second, does scene-context adaptation help small or visually ambiguous categories without hurting large, distinctive objects? Third, does VLM-assisted purification improve pseudo-label quality enough to justify its cost? Fourth, does Gumbel routing add measurable value once prompt hierarchy and purification are already present?

Failure analysis should focus on class pairs that are naturally confusing in aerial imagery. Examples include small vehicle versus large vehicle, ship versus harbor structure, bridge versus road segment, and sports-field subclasses. For each pair, the evaluation should compare class-wise AP, false-positive sources, and prompt sensitivity. Qualitative figures should show accepted and rejected pseudo labels, but the final paper should avoid cherry-picking: rejection examples are as important as success cases because they explain what the VLM purification stage is actually doing.

5 Expected Contributions

The expected contribution is a method and evaluation framework rather than a claim of completed empirical superiority. The first contribution is a hierarchy-aware prompt bank for remote-sensing rotated detection, designed around parent categories, aliases, contrast prompts, and scene-context descriptions. The second is a scene-context prompt adapter that modulates prompt embeddings for dense aerial tiles and small objects. The third is a VLM-assisted pseudo-label purification protocol that adapts the core idea of VLM-purified remote-sensing learning to rotated detection. The fourth is a controlled evaluation plan that treats OpenPrompt/OpenRSD as a baseline scaffold and tests optional Gumbel routing only after the main components are validated.

This framing is intentionally conservative. It avoids claiming final benchmark numbers before experiments are run, and it separates the infrastructure contribution from the research hypothesis. If the method succeeds, the evidence should appear in improved rotated mAP, better class-wise robustness under prompt variants, fewer high-confidence confusion errors, and a cleaner pseudo-label pool.

6 Conclusion

We presented a short method paper for VLM-guided, hierarchy-aware prompt learning in remote-sensing object detection. The proposed GeoNexus-RSD direction updates an older OpenPrompt/OpenRSD-centered plan into a 2026-facing story built around multimodal prompting, taxonomy-aware semantics, scene-context adaptation, and VLM-assisted pseudo-label purification. DOTA-style rotated detection is the main task, while optional Gumbel routing remains a lightweight extension rather than the core claim. The next step is to implement the staged ablations in the local OpenPrompt scaffold and report measured results only after the supervised detector, prompt adapter, and purification pipeline are independently verified.

References

- [1] Huang et al. OpenRSD: Towards Open-prompts for Object Detection in Remote Sensing Images. In *ICCV*, 2025.
- [2] RS-MPOD authors. RS-MPOD: Multimodal Prompting for Remote Sensing Object Detection. arXiv:2602.01954, 2026.
- [3] SemiEarth authors. SemiEarth: VLM-Purified Semi-Supervised Learning for Remote Sensing Semantic Segmentation. arXiv:2602.00202, 2026.
- [4] Alec Radford et al. Learning Transferable Visual Models From Natural Language Supervision. In *ICML*, 2021.
- [5] Edward J. Hu et al. LoRA: Low-Rank Adaptation of Large Language Models. arXiv:2106.09685, 2021.
- [6] Eric Jang, Shixiang Gu, and Ben Poole. Categorical Reparameterization with Gumbel-Softmax. In *ICLR*, 2017.
- [7] Chris J. Maddison, Andriy Mnih, and Yee Whye Teh. The Concrete Distribution: A Continuous Relaxation of Discrete Random Variables. In *ICLR*, 2017.